Lecture 3

Registers: the state of a part

Objectives. *Lecture 2 read a whole register, and only at the end of a circuit. This lecture reads part of one, and asks:* what is the state of part of a register? *The answer, the reduced state (Theorem 3.9), is the one matrix that gives every measurement of the part the statistics the whole register gives; it has a formula in any basis. For the Bell state the answer is* $\frac{1}{2}\mathbb{1}$*, which no ket describes.*

By the end of this lecture you will be able to:

- Build a register from its parts as a tensor product, and decide whether a state of two qubits is a product.
- Compute the probabilities and the states left by a reading of part of a register, and by a measurement in any basis, drawn as a circuit.
- Prove that a global phase changes nothing in any procedure, while a relative phase can be seen.
- Define the reduced state by its property, prove it unique, derive its formula, and compute it for the Bell state.

3.1 Tensor structure and entanglement

Let a register AB of $m+n$ qubits have the part A on its top m wires and the part B on the n wires below. For columns $\psi \in \mathbb{C}^{2^m}$ and $\varphi \in \mathbb{C}^{2^n}$, the Kronecker product of Definition 1.2 is the column $\psi \otimes \varphi \in \mathbb{C}^{2^{m+n}}$ with the entries

$$(\psi \otimes \varphi)_{xx'} = \psi_x \varphi_{x'},$$

one for each string xx' of AB; it is linear in each factor, and is also written $|\psi\rangle\,|\varphi\rangle$. By the interchange law and (2.3),

$$\begin{aligned} \langle \psi \otimes \varphi | \psi' \otimes \varphi' \rangle &= (\psi^\dagger \otimes \varphi^\dagger)(\psi' \otimes \varphi') \\ &= \langle \psi | \psi' \rangle \, \langle \varphi | \varphi' \rangle \,, \end{aligned} \tag{3.1}$$

so a product of states is a state, and $(U \otimes V)(\psi \otimes \varphi) = U\psi \otimes V\varphi$: a gate on one part acts on its own factor. The columns $|x\rangle\,|x'\rangle = |xx'\rangle$ are a basis, by (1.3), so every state of AB is $\sum_{x,x'} c_{xx'}\,|x\rangle\,|x'\rangle$. In the language of *Linear Algebra* (L12), $\mathbb{C}^{2^{m+n}}$ is the tensor product $\mathbb{C}^{2^m} \otimes \mathbb{C}^{2^n}$, realised by Kronecker products. Not every state of AB is a product.

Definition 3.1 (Product, entangled)**.** A state of AB is a *product* if it is $\psi \otimes \varphi$ for a state ψ of A and a state φ of B; otherwise it is *entangled*.

Proposition 3.2 (A test for two qubits)**.** *A state* $\sum_{j,k} c_{jk}\,|jk\rangle$ *of two qubits is a product if and only if* $c_{00}c_{11} = c_{01}c_{10}$.

Proof. If it is $\psi\otimes\varphi$, then $c_{jk}=\psi_j\varphi_k$, and both sides are $\psi_0\psi_1\varphi_0\varphi_1$. Conversely, the condition says that the 2×2 matrix (c_{jk}), which is not zero, has determinant zero, so it has rank one: its rows are $\psi_0 u^{\mathsf{T}}$ and $\psi_1 u^{\mathsf{T}}$ for one non-zero column u. Then $c_{jk}=\psi_j u_k$, and the state is $\psi\otimes u$. By (3.1), $\|\psi\|\,\|u\|=1$, so it is the product of the states $\psi/\|\psi\|$ and $\|\psi\|\,u$. ∎

Example 3.3 (QQ: a product, and an entangled state)**.** The state

$$|+\rangle\,|+\rangle=\tfrac{1}{2}\big(|00\rangle+|01\rangle+|10\rangle+|11\rangle\big)$$

has $c_{00}c_{11}=\frac{1}{4}=c_{01}c_{10}$: it is a product, although every string appears in it. The Bell state

$$\left|\Phi^{+}\right\rangle=\frac{|00\rangle+|11\rangle}{\sqrt{2}}$$

has $c_{00}c_{11}=\frac{1}{2}$ and $c_{01}c_{10}=0$: it is entangled.

QM-I, Lecture 10, uses the same definition of entanglement: not a product. A superposition of every string can still be a product.

3.2 Measuring part of a register

Definition 2.1 defined reading a whole register as measuring an observable with 2^n distinct eigenvalues, each of multiplicity one. A reading of the part A has degenerate outcomes: the outcome k is shared by every string kx'. QM-I, Lecture 4, already uses Lüders' rule for such outcomes. It projects onto the whole eigenspace, rather than choosing one vector in it. We will apply the same rule to a register's wires.

Definition 3.4 (Projective measurement, Lüders' rule)**.** A *projective measurement* of a register is a family of matrices $\Pi_1,\dots,\Pi_K$ with $\Pi_k^\dagger=\Pi_k$, $\Pi_j\Pi_k=\delta_{jk}\Pi_k$ and $\sum_k\Pi_k=\mathbb{1}$. On the state ψ it gives the outcome k with probability

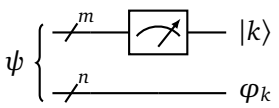
$$p_k=\langle\psi|\,\Pi_k\,|\psi\rangle=\|\Pi_k\psi\|^2\,,$$

and, if $p_k>0$, it leaves the state $\Pi_k\psi/\sqrt{p_k}$ (*Lüders' rule*).

The probabilities add up to $\langle\psi|\psi\rangle=1$. Reading the whole register is the case $\Pi_x=|x\rangle\langle x|$: it gives x with probability $|\psi_x|^2$, as in Lecture 2, and leaves $(\psi_x/|\psi_x|)\,|x\rangle$, which is $|x\rangle$ up to a factor of modulus one; Theorem 3.7 shows that such a factor changes nothing. Reading A is the measurement $\Pi_k=|k\rangle\langle k|\otimes\mathbb{1}$, for $k\in\{0,1\}^m$. On $\psi=\sum_{k,x'}c_{kx'}\,|k\rangle\,|x'\rangle$ it gives

$$\begin{aligned} p_k&=\textstyle\sum_{x'}|c_{kx'}|^2\,,\\ \Pi_k\psi/\sqrt{p_k}&=|k\rangle\otimes\varphi_k,\\ \varphi_k&=p_k^{-1/2}\textstyle\sum_{x'}c_{kx'}\,|x'\rangle\,, \end{aligned}\tag{3.2}$$

drawn as a meter on the wires of A alone, which carry $|k\rangle$ on:



Reading B afterwards gives x' with probability $|c_{kx'}|^2/p_k$, so the two readings give the pair (k,x') with probability $|c_{kx'}|^2$, as one reading of the whole register does.

Example 3.5 (QQ: reading one qubit)**.** Read the top qubit of $|+\rangle|+\rangle$. The outcome 0 has probability $\frac{1}{2}$, and its eigenspace, spanned by $|00\rangle$ and $|01\rangle$, has dimension two; Lüders' rule leaves $|0\rangle|+\rangle$, and the qubit not read is as it was. Read the top qubit of Φ^+: it gives 0 or 1 with probability $\frac{1}{2}$ each, and leaves $|00\rangle$ or $|11\rangle$. The qubit not read now holds $|k\rangle$, and a reading of it gives k with certainty.

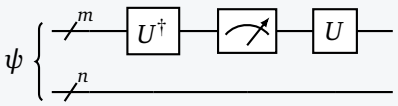
The word "collapse" names this update. It is not a dynamics: it is the rule by which the predictions for later steps are conditioned on the outcome.

Other bases. For a gate U on A, the basis $\{U|k\rangle\}$ of A (every orthonormal basis is one) gives the projective measurement

$$\Pi_k^U = U|k\rangle\langle k|U^\dagger \otimes \mathbb{1},$$

the measurement of A *in the basis* $\{U|k\rangle\}$.

Proposition 3.6 (A measurement in a basis, as a circuit)**.** *In the circuit*



(i) $U^\dagger$ and the reading of A give k with the probability $\left\|\Pi_k^U\psi\right\|^2$;

(ii) the reading leaves A in $|k\rangle$, and U applied after it leaves the Lüders state $\Pi_k^U\psi/\left\|\Pi_k^U\psi\right\|$.

Proof. By the interchange law, $\Pi_k^U = (U\otimes\mathbb{1})(|k\rangle\langle k|\otimes\mathbb{1})(U^\dagger\otimes\mathbb{1})$. Since $U\otimes\mathbb{1}$ is unitary, $\left\|\Pi_k^U\psi\right\|$ is the norm of $(|k\rangle\langle k|\otimes\mathbb{1})(U^\dagger\otimes\mathbb{1})\psi$, whose square is the probability of reading k after $U^\dagger$: this is (i). By (3.2) the reading leaves A in $|k\rangle$, and $U\otimes\mathbb{1}$ turns the state it leaves into $\Pi_k^U\psi$ divided by the same norm: this is (ii). ∎

So $U^\dagger$ and a reading give the probabilities of the measurement in the basis $\{U|k\rangle\}$, but leave $|k\rangle$, not $U|k\rangle$. For one qubit and $U = H$ the basis is $|\pm\rangle$, the eigenvectors of X. Nielsen and Chuang (p. 22) say that this measurement leaves $|+\rangle$ or $|-\rangle$, which is the Lüders state; their recipe of Ex. 4.33, a change of basis and then a reading, gives the circuit of H and a reading, which leaves $|0\rangle$ or $|1\rangle$. In this course, "measure in the basis $\{U|k\rangle\}$" means the projective measurement, and its circuit is $U^\dagger$, a reading, and U.

3.3 Phases

Lecture 2 promised that the sign $(-1)^{f(0)}$ left by Deutsch's circuit (Example 2.6) changes nothing, whatever follows. A *procedure* is a sequence of steps, each a gate, the joining of a fresh register in a given state, or a projective measurement; each step may be chosen according to the outcomes before it.

Theorem 3.7 (Global and relative phases)**.** *Let ψ be a state and α real.*

(i) For every procedure, the runs on ψ and on $e^{i\alpha}\psi$ give every sequence of outcomes with the same probability, and so does a run in which a gate U of the procedure is replaced by $e^{i\alpha}U$.

(ii) $|+\rangle$ and $|-\rangle$, read, give 0 and 1 with probability $\frac{1}{2}$ each; H and a reading tell them apart with certainty.

Proof. (i) We argue by induction on the number of steps. Every sequence h of outcomes so far has the same probability in the two runs. If this probability is positive, the states it leaves are ψ_h and $c_h\psi_h$, with $|c_h| = 1$. At the start, c is $e^{i\alpha}$, or 1 when a gate is replaced. The next step depends only on h, so it is the same in both runs.

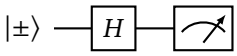
A gate sends $c_h\psi_h$ to $c_hU\psi_h$. Replacing it by $e^{i\alpha}U$ gives $e^{i\alpha}c_hU\psi_h$, still a multiple of modulus one.

Joining a register in state φ gives $c_h(\psi_h \otimes \varphi)$.

A projective measurement gives k with probability $\|\Pi_k c_h\psi_h\|^2 = \|\Pi_k\psi_h\|^2$. If this is positive, it leaves c_h times the state obtained from ψ_h. Thus every step preserves the induction claim.

(ii) The amplitudes of $|\pm\rangle$ are $\pm 1/\sqrt{2}$, and $H|+\rangle = |0\rangle$, $H|-\rangle = |1\rangle$ by (2.2), since $H^2 = \mathbb{1}$. ∎

The predictions depend on a state only through its *ray*, the set of its multiples $e^{i\alpha}\psi$, and the sign in Example 2.6 may be dropped. A *relative* phase, the sign of one amplitude of $|-\rangle$, is seen by (ii):



By (i), a gate may be replaced by $e^{i\alpha}$ times itself, as $U \simeq V$ (equal up to a global phase) allows; inside a controlled gate it may not, since there the phase becomes relative (Lecture 6).

3.4 Density operators recalled, the reduced state

Density operators, recalled from QM-I, Lecture 3. The density operator of a state ψ is $|\psi\rangle\langle\psi|$. If the state ψ_i is prepared with probability p_i, the *mixture* has the density operator

$$\rho = \textstyle\sum_i p_i |\psi_i\rangle\langle\psi_i| ,$$

as in QM-I. Since $\langle\psi|\Pi|\psi\rangle = \mathrm{Tr}(\Pi|\psi\rangle\langle\psi|)$, averaging over i the rules for gates and for projective measurements gives them in terms of ρ alone: a gate U takes ρ to $U\rho U^\dagger$, and a projective measurement gives

$$\begin{aligned} p_k &= \mathrm{Tr}(\rho\Pi_k), \\ \rho &\mapsto \Pi_k\rho\Pi_k/p_k. \end{aligned} \tag{3.3}$$

The update is Lüders' rule averaged by Bayes' rule: given k, with $p_k > 0$, each ψ_i with $\Pi_k\psi_i \neq 0$ has the probability $p_i\langle\psi_i|\Pi_k|\psi_i\rangle/p_k$ and has become $\Pi_k\psi_i/\|\Pi_k\psi_i\|$, while the others have probability 0 and contribute nothing to $\Pi_k\rho\Pi_k$. The expectation of an observable M is $\mathrm{Tr}(\rho M)$, as in QM-I. For a pure state, the density operator represents its ray (Theorem 3.7).

Definition 3.8 (Density operator). A *density operator* on $\mathbb{C}^d$ is a self-adjoint $d \times d$ matrix ρ with $\langle v|\rho|v\rangle \geq 0$ for every v and $\mathrm{Tr}\,\rho = 1$. It is *pure* if $\rho = |\psi\rangle\langle\psi|$ for a state ψ, and *mixed* otherwise.

We also call density operators *states*. Pure states have unit-vector descriptions; mixed states do not.

Every mixture is one, since $\mathrm{Tr}\,\rho = \sum_i p_i = 1$ and $\langle v|\rho|v\rangle = \sum_i p_i |\langle\psi_i|v\rangle|^2$. Conversely, by the spectral theorem (*Linear Algebra*, L10), $\rho = \sum_j \lambda_j |e_j\rangle\langle e_j|$ with the e_j orthonormal, and the $\lambda_j = \langle e_j|\rho|e_j\rangle \geq 0$ add up to $\mathrm{Tr}\,\rho = 1$: every density operator is a mixture. A pure $|\psi\rangle\langle\psi|$ has the eigenvalue 1, on ψ, and 0 on the vectors orthogonal to ψ.

The state of a part. QM-I, Lecture 11, characterises a reduced state by all local measurements. We will use that characterisation for registers. Let AB be in the state ρ. A projective

measurement $\{\Pi_k\}$ of A alone is the measurement $\{\Pi_k \otimes \mathbb{1}\}$ of AB, with the probabilities $\operatorname{Tr}(\rho(\Pi_k \otimes \mathbb{1}))$. A state of A should predict these, and the expectation of every observable of A, as the state of the whole does.

Theorem 3.9 (The reduced state). *Let ρ be a density operator on $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$.*

(i) There is exactly one $d_A \times d_A$ matrix ρ_A with

$$\operatorname{Tr}(\rho_A M) = \operatorname{Tr}\big(\rho(M \otimes \mathbb{1})\big) \tag{3.4}$$

for every $d_A \times d_A$ matrix M. It is the reduced state *of A, written $\operatorname{Tr}_B \rho$.*

(ii) For every orthonormal basis $f_1, \dots, f_{d_B}$ of $\mathbb{C}^{d_B}$,

$$\rho_A = \textstyle\sum_b \iota_b^\dagger \rho \iota_b, \tag{3.5}$$

where the map $\iota_b \colon \mathbb{C}^{d_A} \to \mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$ appends f_b:

$$\iota_b v = v \otimes f_b.$$

Its matrix is $\iota_b = \mathbb{1} \otimes |f_b\rangle$, the Kronecker product of the $d_A \times d_A$ identity and the column f_b. In the computational basis, $(\rho_A)_{aa'} = \sum_b \langle ab| \rho |a'b\rangle$.

(iii) ρ_A is a density operator.

Proof. (i) If σ and σ' both satisfy (3.4), then $D = \sigma - \sigma'$ has $\operatorname{Tr}(DM) = 0$ for every M. For $M = |a'\rangle\langle a|$ this says $\langle a| D |a'\rangle = 0$, so $D = 0$.

(ii) By (2.3), $\iota_b^\dagger = \mathbb{1} \otimes \langle f_b|$, and by the interchange law $\iota_b M = M \otimes |f_b\rangle = (M \otimes \mathbb{1})\iota_b$. Let σ be the right side of (3.5). Since $\operatorname{Tr}(XY) = \operatorname{Tr}(YX)$ for a $p \times q$ matrix X and a $q \times p$ matrix Y,

$$\begin{aligned} \operatorname{Tr}(\sigma M) &= \textstyle\sum_b \operatorname{Tr}\big(\rho \iota_b M \iota_b^\dagger\big) \\ &= \textstyle\sum_b \operatorname{Tr}\big(\rho(M \otimes \mathbb{1})\iota_b \iota_b^\dagger\big) \\ &= \operatorname{Tr}\big(\rho(M \otimes \mathbb{1})\big), \end{aligned}$$

because $\sum_b \iota_b \iota_b^\dagger = \mathbb{1} \otimes \sum_b |f_b\rangle\langle f_b| = \mathbb{1}$. So σ satisfies (3.4), which proves existence, and $\sigma = \rho_A$ by (i), whatever the basis. For $f_b = |b\rangle$, $\iota_b |a'\rangle = |a'b\rangle$ gives the entries.

(iii) $M = \mathbb{1}$ gives $\operatorname{Tr}\rho_A = \operatorname{Tr}\rho = 1$. Each term $\iota_b^\dagger \rho \iota_b$ is self-adjoint, and $\langle v| \rho_A |v\rangle = \sum_b \langle v \otimes f_b| \rho |v \otimes f_b\rangle \geq 0$, since $\iota_b v = v \otimes f_b$. ∎

Both sides of (3.4) are linear in M, and every matrix is $M_1 + iM_2$ with $M_1 = (M + M^\dagger)/2$ and $M_2 = (M - M^\dagger)/2i$ self-adjoint; so asking (3.4) of every observable is asking it of every matrix. One observable is not enough: $\frac{1}{2}\mathbb{1}$ and $|+\rangle\langle +|$ give the reading in Z the same probabilities, $\frac{1}{2}$ and $\frac{1}{2}$, while $\operatorname{Tr}(\frac{1}{2}X) = 0$ and $\langle +| X |+\rangle = 1$. The state $\rho_B = \operatorname{Tr}_A \rho$ of B is defined in the same way, with $\mathbb{1} \otimes M$.

Corollary 3.10 (A part of a pure state). *Let $f_1, \dots, f_{d_B}$ be an orthonormal basis of $\mathbb{C}^{d_B}$, and let χ be a state of AB. Then $\chi = \sum_b \chi_b \otimes f_b$ with $\chi_b = \iota_b^\dagger \chi$, and*

$$\operatorname{Tr}_B |\chi\rangle\langle\chi| = \textstyle\sum_b |\chi_b\rangle\langle\chi_b| .$$

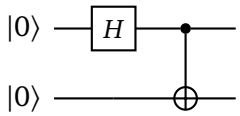
In particular, the part A of a product $\psi \otimes \varphi$ is in the pure state $|\psi\rangle\langle\psi|$.

Proof. Since $\sum_b \iota_b \iota_b^\dagger = \mathbb{1}$, $\chi = \sum_b \iota_b \chi_b = \sum_b \chi_b \otimes f_b$, and the terms of (3.5) for $\rho = |\chi\rangle\langle\chi|$ are $\iota_b^\dagger |\chi\rangle\langle\chi| \iota_b = |\chi_b\rangle\langle\chi_b|$. For the product, take a basis with $f_1 = \varphi$: by the interchange law $\iota_b^\dagger(\psi \otimes \varphi)$ is $\psi \langle f_b|\varphi\rangle$, which is ψ for $b = 1$ and 0 otherwise. ∎

Measuring B in the basis $\{f_b\}$ is the projective measurement $\Pi_b = \iota_b \iota_b^\dagger = \mathbb{1} \otimes |f_b\rangle\langle f_b|$, and $\Pi_b \chi = \chi_b \otimes f_b$: it gives b with probability $\|\chi_b\|^2$ and, if that is positive, leaves A in $\chi_b/\|\chi_b\|$. So $\mathrm{Tr}_B |\chi\rangle\langle\chi|$ is the mixture of these states with these probabilities, and it is the same for every basis of B.

3.5 The Bell state

Example 3.11 (QQ: the Bell state and its half)**.** A Hadamard, then a CNOT, makes Φ^+ (Nielsen and Chuang, §1.3.6):



$$= |\Phi^+\rangle \tag{3.6}$$

an equality of vectors: $(H \otimes \mathbb{1})|00\rangle = |+\rangle|0\rangle$, which is $(|00\rangle + |10\rangle)/\sqrt{2}$, and the CNOT turns $|10\rangle$ into $|11\rangle$. In the computational basis of B, $\Phi^+ = (|0\rangle/\sqrt{2}) \otimes |0\rangle + (|1\rangle/\sqrt{2}) \otimes |1\rangle$, so by Corollary 3.10

$$\begin{aligned}\mathrm{Tr}_B |\Phi^+\rangle\langle\Phi^+| &= \tfrac{1}{2}|0\rangle\langle 0| + \tfrac{1}{2}|1\rangle\langle 1| \\ &= \tfrac{1}{2}\mathbb{1},\end{aligned}$$

and $\mathrm{Tr}_A |\Phi^+\rangle\langle\Phi^+| = \frac{1}{2}\mathbb{1}$ in the same way.

The half has no ket. If a state ψ of one qubit predicted every measurement of A, so that, for $\rho = |\Phi^+\rangle\langle\Phi^+|$, $\mathrm{Tr}(\rho(M \otimes \mathbb{1})) = \langle\psi| M |\psi\rangle = \mathrm{Tr}(|\psi\rangle\langle\psi| M)$ for every M, then by the uniqueness in Theorem 3.9, $\frac{1}{2}\mathbb{1} = |\psi\rangle\langle\psi|$; but $|\psi\rangle\langle\psi|$ has the eigenvalue 1, and $\frac{1}{2}\mathbb{1}$ has only $\frac{1}{2}$. In every basis $\{U|k\rangle\}$ the half gives each outcome with probability $\mathrm{Tr}(\frac{1}{2}U|k\rangle\langle k|U^\dagger) = \frac{1}{2}$, as its reading in Example 3.5 did. In Z this matches $|+\rangle$; in X it does not, so $|+\rangle$ is not the half's state. The pair is in a pure state, yet its half has no ket: the state of a part is $\mathrm{Tr}_B \rho$.

Summary.

- *Handed on:* registers as tensor products, and entangled as not a product (Definition 3.1 and Proposition 3.2); Lüders' rule for reading a part, and the measurement in the basis $\{U|k\rangle\}$ as $U^\dagger$, a reading and U (Definition 3.4 and Proposition 3.6).
- *Phases:* a global phase changes nothing in any procedure; a relative one can be seen (Theorem 3.7).
- *The main result:* the state of a part is $\mathrm{Tr}_B \rho$, the one matrix that predicts every measurement of the part, with a formula in every basis (Theorem 3.9 and Corollary 3.10).
- *QQ:* the Bell state is pure, and its halves are in $\frac{1}{2}\mathbb{1}$, which no ket describes (Example 3.11).
- *Next:* Lecture 4, what can be done with an unknown state.

References

QM-I, Lectures 3, 4, 10 and 11. Nielsen and Chuang, pp. 16–17, 22, 25–26, 71–74, 87–88, 93–96, 99–107 (§2.4; Box 2.6, p. 107) and 187–188 (Ex. 4.33). Watrous (2025), pp. 43–53 (little-endian), 85–86 and 279–283. Kitaev, Shen and Vyalyi, Def. 10.2, p. 96. G. Lüders, Ann. Phys. (Leipzig) **8** (1951), 322–328.